

Impact of acute caffeine ingestion on endothelial function in subjects with and without coronary artery disease.



Although coffee is a widely used, pharmacologically active beverage, its impact on the cardiovascular system is controversial. To explore the effect of acute caffeine ingestion on brachial artery flow-mediated dilation (FMD) in subjects without coronary artery disease (CAD; controls) and patients with CAD, we prospectively assessed brachial artery FMD in 40 controls and 40 age- and gender-matched patients with documented stable CAD on 2 separate mornings 1 week to 2 weeks apart. After overnight fasting, discontinuation of all medications for ≥ 12 hours, and absence of caffeine for >48 hours, participants received capsules with caffeine 200 mg or placebo. One hour after drug ingestion, participants underwent brachial artery FMD and nitroglycerin-mediated dilation (NTG) using high-resolution ultrasound. As expected, patients with CAD were more often diabetic, hypertensive, obese, dyslipidemic, and smoked more than controls ($p < 0.01$ for all comparisons). Aspirin, Clopidogrel, angiotensin-converting enzyme inhibitors, β blockers, and statins were significantly more common in patients with CAD than in controls ($p < 0.01$ for all comparisons). At baseline, FMD, but not NTG, was significantly lower in patients with CAD compared to controls. Acute caffeine ingestion significantly increased FMD (patients with CAD $5.6 \pm 5.0\%$ vs $14.6 \pm 5.0\%$, controls $8.4 \pm 2.9\%$ vs $18.6 \pm 6.8\%$, $p < 0.001$ for all comparisons) but not NTG (patients with CAD $13.0 \pm 5.2\%$ vs $13.8 \pm 6.1\%$, controls $12.9 \pm 3.9\%$ vs $13.9 \pm 5.8\%$, $p = \text{NS}$ for all comparisons) and significantly decreased high-sensitivity C-reactive protein (patients with CAD 2.6 ± 1.4 vs 1.4 ± 1.2 mg/L, controls 3.4 ± 3.0 vs 1.2 ± 1.0 mg/L, $p < 0.001$ for all comparisons) in the 2 groups compared to placebo. In conclusion, acute caffeine ingestion significantly improved endothelial function assessed by brachial artery FMD in subjects with and without CAD and was associated with lower plasma markers of inflammation. Copyright © 2011 Elsevier Inc. All rights reserved.